



LOREN

DVB-H Receiver IC

Overview

This DVB-H Receiver IC is based on SIDSA's DVB-H Intellectual Property block.

This complete System On Chip includes a programmable RF interface with dual high speed ADC, high performance DVB-H demodulator, MPE-FEC, Transport Stream Processor and an ARM946™ microcontroller unit for running customer code and DVB-H software stack.

This device also features a very rich set of different interfaces that are common in mobile or handheld devices, such as SDIO, Compact Flash, USB and even Ethernet.

One key feature of this device is its support for a wide range of different IF and base band DVB-H tuners. Its programmable RF front-end provides the flexibility needed to evolve and expand this coverage in the future.

Usage Model

This IC is available as a demonstrator of the IP and is the core of our professional receiver, RF support and IP evaluation kits.

SIDSA also licenses system-level IP for different types of receivers such as SDIO, Compact Flash and USB modules. These modules can be licensed by manufacturing companies and are ideal for embedding inside receivers, Set-Top-Boxes, PC's or PDA's.

Software

SIDSA provides a complete DVB-H software stack to ease the integration of our licensee's ICs in any target application.

SIDSA follows in its software a consistent approach of using Internet Protocol based standards in its solution.

The drivers for the different physical interfaces such as SDIO and CompactFlash appear to the host OS as pure Ethernet devices. RPC is used for control. Service discovery is done internally by parsing PSI/SI and ESG and announced using SAP. IGMP is then used for joining the services.

This layered, modular approach abstracts all the DVB-H complexity and greatly facilitates the integration with the host.

DVB-T/H Hardware

- ◆ Supports both Intermediate Frequency and base band Tuners
- ◆ Programmable tuner interface
- ◆ Up to 110 Hz Doppler in 16-QAM, GI 1/4, CR 2/3
- ◆ 2k, 4k and 8k modes
- ◆ 8 MHz, 7 MHz, 6 MHz and 5 MHz
- ◆ QPSK, 16-QAM and 64-QAM
- ◆ Guard interval options: 1/32, 1/16, 1/8 and 1/4
- ◆ Hierarchical modulation support
- ◆ Transport Stream Processor
- ◆ MPE-FEC and time slicing
- ◆ Multi-service reception limited only by external memory size
- ◆ Advanced power management with fast re-synchronization

ARM946 Embedded Processor

- ◆ ARM946™ MCU
- ◆ Booting is possible from different host interfaces.
- ◆ Internal RAM for program or data.
- ◆ External memory interface for MPE FEC memory
- ◆ Optional external program Flash

Interfaces

- ◆ SDIO Card
- ◆ Compact Flash
- ◆ USB 1.1
- ◆ Ethernet
- ◆ Transport Stream (DVB-T or re-encapsulated)
- ◆ Generic Microprocessor Slave Interface
- ◆ I2C, UART, GPIO
- ◆ Smart Card

External description

System control

Clocking, reset, software debug access and boot configuration.

Tuner interfaces

Used to configure and control the RF tuner and to interface with its analog output, either in base-band or intermediate frequency. These include:

- ◆ Analog inputs: I & Q and IF
- ◆ I2C controllers (2x)
- ◆ PWM outputs for gain control (2x)
- ◆ Generic tuner control I/Os.

Host interfaces:

The demodulated and corrected information is sent to the host in charge of video decoding and presentation.

- ◆ SDIO

- ◆ Compact Flash
- ◆ USB1.1 Full-Speed
- ◆ Ethernet 10/100 Mbps
- ◆ Transport Stream: pre MPE-FEC or post MPE-FEC
- ◆ Generic microprocessor bus slave

Memory interfaces

External memory is used for the FEC correction process on DVB-H data and can also be used to store program and data for the internal ARM microprocessor. The IC supports the following types of memories on the same bus: SDRAM, SRAM and Flash. Firmware is also loaded from an external I2C memory.

Auxiliary interfaces

UART for console access, smart card interface for secure features, external inputs for interrupt and DMA requests and general purpose pins.

